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## Planetary Alignment Round 1 — DeepSeek Packet: Report

Date: 2025-07-10

Model/version: DeepSeek

Live search: Not available in this session

Scope: Literature and available-data inquiry only. No fieldwork, intervention, outreach, or claim of contact or authorization.

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### 1. Scope and Access

Search terms: information theory of individuality; plant–fungal information transfer; mycorrhizal network signaling; biological individuality ontology; Gaia hypothesis atmospheric homeostasis; reciprocal rewards mycorrhizal symbiosis; holobiont individuality; fungal decision-making; plant intelligence information processing.

Sources inspected: Krakauer et al. (2020) — full abstract and PubMed record; Lovelock & Margulis (1974) — full text via jameslovelock.org; Kiers et al. (2011) — abstract and metadata via Science; additional sources on mycorrhizal individuality, common mycorrhizal networks, process ontology, fungal signaling, and plant cognition retrieved through search.

Access limits: Several key sources were accessible only as abstracts or metadata (Kiers et al.; some ScienceDirect chapters). The full text of Krakauer et al. (2020) was available via PMC. No primary data collection was performed.

Significant omissions: Non-English literature on plant–fungal relations; Indigenous and traditional ecological knowledge frameworks; detailed mathematical treatments of information decomposition beyond what abstracts describe.

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### 2. Three Priority Claims or Open Questions

# Proposition Source & Location Evidence Type Boundary & Timescale Alternatives Support / Weaken / Unresolved

1 Mycorrhizal collectives (plant + fungus) qualify as evolutionary individuals under a graded Darwinian individuality model Philosophy of Science Archive preprint on mycorrhizal individuality

(metadata not fully verified in this session) Conceptual argument; theoretical individuation Single plant–fungus pair; evolutionary timescale Holobiont scepticism (Godfrey-Smith, Skillings); Hull's interactor model Support: pseudo-vertical transmission demonstrated. Weaken: horizontal acquisition of symbionts complicates parent–offspring lineages. Unresolved: degree of individuality varies; requires probabilistic model

2 Common mycorrhizal networks transport defense signals between plants, constituting a form of interplant communication Reviews on plant–plant communication through common mycorrhizal networks (e.g., *Advances in Botanical Research*; access abstract-only) Experimental review; laboratory and limited field evidence Multiple plants connected by AM/ECM fungi; ecological timescale (hours–days) Signal transfer may be passive diffusion rather than active fungal control; mechanism unknown Support: measurable protection benefits. Weaken: few field studies; difficulty identifying CMNs in situ. Unresolved: nature of signals and degree of fungal control

3 The Gaia hypothesis posits that the biosphere regulates planetary conditions through feedback mechanisms Lovelock & Margulis (1974), *Tellus XXVI* Conceptual argument; atmospheric chemistry evidence Planetary scale; geological timescale ( $10^9$  years) Coincidental homeostasis; inorganic equilibrium processes; Neo-Darwinian critique (Dawkins) Support: anomalous atmospheric disequilibrium ( $O_2 + CH_4$ ). Weaken: difficulty distinguishing biological from abiotic regulation. Unresolved: testable predictions at planetary scale remain contested

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### 3. Disagreement Map

#### Empirical disagreements:

- CMN function: Whether common mycorrhizal networks genuinely facilitate directed signaling or merely provide passive diffusion pathways remains disputed. The mechanism of signal transfer is explicitly described as unknown.
- Fungal cognition: Some researchers attribute decision-making, learning, and memory to fungal mycelia, while others caution against anthropomorphic projection. The debate turns on what counts as evidence for cognition without a nervous system.
- Plant intelligence: Definitions vary. Some equate vascular systems with information processing; others require problem-solving capacity toward a goal.

#### Conceptual disagreements:

- Individuality criteria: Organismal individuality (bounded body, single genome) versus relational or processual individuality (emergent from interaction patterns). The ITI explicitly expands allowable individuals to include distributed aggregations without physical boundaries.
- Gaia hypothesis: Whether planetary homeostasis is an adaptation “by and for” the biosphere or an emergent consequence of local interactions remains a live conceptual dispute.

Detection bias and under-recognition: Methods suited to detecting organismal individuality (genomic sequencing, physical boundaries) may systematically miss colonial or driven individuality that the ITI predicts. Conversely, the risk of over-attribution exists when signals are interpreted as intentional communication without ruling out passive transport.

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#### 4. Ontology and Boundary Note

Assumed ontology: This analysis assumes a relational–processual ontology, in which individuals are not substances with fixed boundaries but temporary stabilizations of interaction patterns. This aligns with the ITI’s Gestalt approach, where figure–ground distinctions are made by selection and observation.

Alternative account: Mechanistic–organismal ontology treats individuals as discrete entities with clear boundaries, typically defined by a single genome or physical membrane. This view is implicit in much of mainstream biology.

What each reveals or obscures:

- Relational–processual: Reveals distributed, transient, and nested individuality (e.g., mycorrhizal collectives that overlap). Obscures the practical utility of stable units for measurement and intervention.
- Mechanistic–organismal: Reveals clear units for experimental manipulation. Obscures constitutive relations that cross boundaries (e.g., fungal hyphae integrated into plant root tissue).

Testability: The disagreement can be partially tested by applying information-theoretic measures (e.g., mutual information between system and environment) to mycorrhizal systems. However, the choice of partition and coarse-graining remains a modelling decision, not a purely empirical matter.

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#### 5. Next Bounded Desk-Research Step

Question: Can information-theoretic measures distinguish between active fungal signaling and passive diffusion in common mycorrhizal networks?

What would be needed: A systematic review of existing datasets that report time-series or spatial data on signal transfer in CMNs, combined with a simulation comparing expected information flow under active versus passive transport models. Specifically:

- Partition specification: define donor plant, receiver plant, and fungal network as system components.
- Measure: transfer entropy or directed mutual information.
- Baseline: a null model assuming diffusion alone.
- Confounds: shared forcing (e.g., common environmental stress), autocorrelation, measurement resolution, unequal system size.

Decision rule: If transfer entropy significantly exceeds the diffusion baseline after controlling for confounds, active signaling is supported. If not, the passive hypothesis remains viable. If data are insufficient, the question remains unresolved — a legitimate outcome.

Retain dissent: No consensus is manufactured. The assignment's commitment to preserving unresolved and non-identifiable cases is maintained throughout.

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## References

Krakauer, D. C., Bertschinger, N., Olbrich, E., Flack, J. C., & Ay, N. (2020). The information theory of individuality. *Theory in Biosciences*, 139(2), 209–223.  
<https://doi.org/10.1007/s12064-020-00313-7>

Lovelock, J. E., & Margulis, L. (1974). Atmospheric homeostasis by and for the biosphere: the Gaia hypothesis. *Tellus*, XXVI(1–2), 2–10.

Kiers, E. T., Duhamel, M., Beesetty, Y., et al. (2011). Reciprocal rewards stabilize cooperation in the mycorrhizal symbiosis. *Science*, 333(6044), 880–882.  
<https://doi.org/10.1126/science.1208473>

Additional sources consulted on process ontology, fungal decision-making, plant intelligence, and common mycorrhizal networks. Some source metadata for non-starting sources should be independently verified before citation.

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Note: Cross-model agreement is not independent empirical confirmation; models may share training sources and biases.